Lecture 9

02 February 2026 11:35

Recap

local minima $f: S \to \mathbb{R}$

$\bar{x}$ is called a local minimum iff $\exists\, \epsilon > 0$ s.t.

$f(\bar{x}) \le f(x) \quad \forall x \in B_\epsilon(\bar{x})$

global minima x^* is called a global minimum iff

$f(x^*) \le f(x) \quad \forall x \in S$

Local/global maxima

- similarly defined.

Necessary conditions for local minima

first order conditions

$S \subseteq \mathbb{R}^n$ $(an\ \ open\ \ set)$, $f: S \to \mathbb{R}$

First order necessary condition: ~~If $x^* \in$~~ S is a local minimum (or a local maximum) and f is differentiable at ~~x^* then $\nabla f(x^*) = 0$~~.

- If $\nabla f(x) = 0$ then x is called a stati~~onary~~ point
- A stationary point need not be a local maximum or a local minimum
- If f is a function of a single variable $\nabla f(x)$ is replaced by $f'(x)$

Today

Second order necessary conditions:

- If $x^* \in S$ is a local minimum and f is twice differentiable at x^* then $\nabla^2 f(x^*)$ is positive semidefinite (a symmetric matrix)
- If $x^* \in S$ is a local maximum and f is twice differentiable at x^* then $\nabla^2 f(x^*)$ is negative semidefinite
  - $\nabla^2 f(x^*)$ is positive semidefinite $\not\Rightarrow$ x^* is a local minimum
  - $\nabla^2 f(x^*)$ is negative semidefinite $\not\Rightarrow$ x^* is a local maximum
  - If f is a function of a single variable "$\nabla^2 f(x)$ is positive semidefinite" is replaced by "$f''(x) \ge 0$" and "$\nabla^2 f(x)$ is negative semidefinite" is replaced by "$f''(x) \le 0$"

Examples

① $f(x) = x^2$

$f'(x) = 2x$

$f''(x) = 2$

First order necessary cond. $f'(x) = 0$

$\Rightarrow 2x = 0 \Rightarrow x = 0$

$f''(0) = 2 > 0$ (2nd order condition also satisfied) (for local min)

$\Rightarrow x = 0$ is a local minimum

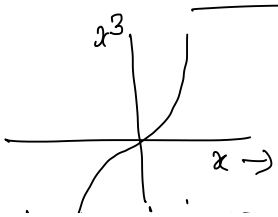
② $f(x) = x^3$

$f'(x) = 3x^2$

$f''(x) = 6x$

First order cond. $f'(x) = 0 \Rightarrow x = 0$

$f''(x)|_{x=0} = 0$ (2nd order necessary condition for local minimum is satisfied)



But $x=0$ is not a local minimum.

How to use the necessary conditions

$f: S \to \mathbb{R}$ (S is open)

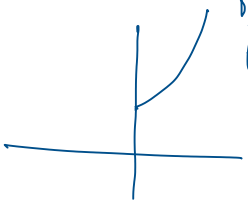
- look for all the points that satisfy both the necessary conditions
- Compare the values of f at these points.

If S is closed

- Consider the boundary points separately.

$f(x) = (x+2)^2$

$f: [0,\infty) \to \mathbb{R}$



A potential approach to find local minima (or local maxima):

- Find all the candidates for local minima (the points that satisfy the first order and the second order necessary conditions)
- Compute f at these points and compare the values to find minima

Example ① $f: \mathbb{R} \to \mathbb{R}$

$f(x) = 2x^3 - 3x^2$ (Find minima and maxima)

$f'(x) = 6x^2 - 6x$

set $f'(x) = 0$

$\Rightarrow 6x(x-1) = 0$

$\Rightarrow x = 0$ or 1

$f''(x) = 12x - 6$

$f''(0) = -6$ ($x=0$ satisfies 2nd order necessary condition for local maxima)

$f''(1) = 6$ ($x=1$ " " " local minima)

$x=0$ is the only candidate for local maxima

$x=1$ " " " minima.

② $f: \mathbb{R}^2 \to \mathbb{R}$

$f(x) = \frac{1}{2}x_1^2 + x_1x_2 + 2x_2^2 - 4x_1 - 4x_2 - x_2^3$

$\nabla f(x) = \begin{bmatrix} x_1 + x_2 - 4 = \\ x_1 + 4x_2 - 4 - 3x_2^2 \end{bmatrix}$

$\nabla f(x) = 0 \Rightarrow x_1 + x_2 = 4$

$x_1 + 4x_2 - 3x_2^2 = 4 \Leftarrow$

Two equations and two unknowns

$x = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$ are the two solutions.

2nd order condition

$\nabla^2 f(x) = \begin{bmatrix} 1 & 1 \\ 1 & 4 - 6x_2 \end{bmatrix}$

$\nabla^2 f\left(\begin{bmatrix} 4 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 & 1 \\ 1 & 4 \end{bmatrix}$ (this is positive definite) (Exercise)

$D^2f\left(\begin{bmatrix}3\\1\end{bmatrix}\right) = \begin{bmatrix}1 & 1\\1 & -2\end{bmatrix}$ (neither PSD nor NSD)

$x = \begin{bmatrix}4\\0\end{bmatrix}$ is the only candidate for local minima.

$x = \begin{bmatrix}3\\1\end{bmatrix}$ can not be a local maximum or a local minimum

$S \subseteq \mathbb{R}^n, \quad f: S \to \mathbb{R}$

Sufficient conditions:

- If f is twice differentiable at $x^* \in S$, $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive definite then $x^* \in S$ is a strict local minimum

  ($f''(x^*) > 0$ if f is a fn. of single variable)

- If f is twice differentiable at $x^* \in S$, $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is negative definite then $x^* \in S$ is a strict local maximum

  ($f''(x) < 0$ if f is fn. of single variable)

Example ① In the above example $x = \begin{bmatrix}4\\0\end{bmatrix}$ is strict local minimum.

② $f: \mathbb{R}^2 \longrightarrow \mathbb{R}$

$f(x) = x_1^3 + x_2^2$

$\nabla f(x) = \begin{bmatrix}3x_1^2\\2x_2\end{bmatrix}$

$\nabla f(x) = 0 \Rightarrow x = \begin{bmatrix}0\\0\end{bmatrix}$

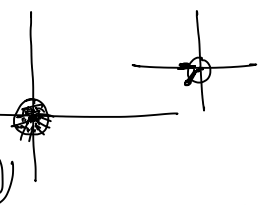
$D^2 f(x) = \begin{bmatrix}6x_1 & 0\\0 & 2\end{bmatrix}$

$D^2 f\left(\begin{bmatrix}0\\0\end{bmatrix}\right) = \begin{bmatrix}0 & 0\\0 & 2\end{bmatrix}$ (Eigenvalues are 0 and 2 $\Rightarrow$ this PSD)

$\begin{bmatrix}0\\0\end{bmatrix}$ is a candidate for local minima or maxima.

$f\left(\begin{bmatrix}0\\0\end{bmatrix}\right) = 0$

$= f\left(\begin{bmatrix}-\epsilon\\-\epsilon\end{bmatrix}\right) = (-\epsilon)^3 + (\epsilon)^2 = \epsilon^2(1-\epsilon) > 0$



$f\left(\begin{bmatrix}-\epsilon\\0\end{bmatrix}\right) = (-\epsilon)^3 + 0 = -\epsilon^3 < 0 = f\left(\begin{bmatrix}0\\0\end{bmatrix}\right)$

$\Rightarrow \begin{bmatrix}0\\0\end{bmatrix}$ is neither a local minimum nor a local maximum

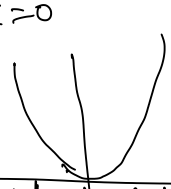
③ $f: \mathbb{R} \longrightarrow \mathbb{R}$

$f(x) = x^4$

$f(x) = 4x^3 \qquad f'(x) = 0 \Rightarrow x = 0$

$f''(x) = 12x^2$

$f''(0) = 0$



Does $x = 0$ satisfy the sufficient conditions for local minima.

But $x = 0$ is a global minimum

Q. Are there problems where necessary conditions are also sufficient & local minima are also global minima.

$$\min_{x \in S} f(x)$$

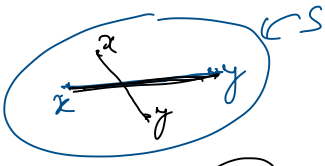
Convex Functions

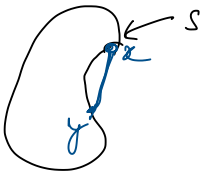
Convex sets

$S \subseteq \mathbb{R}^n$ is called a convex set if $\forall\, x, y \in S$
$\alpha x + (1-\alpha) y \in S \quad \forall\, \alpha \in [0,1]$

$\alpha = 0.5 \qquad \dfrac{x+y}{2}$

$\alpha = 1/3 \qquad \dfrac{x+2y}{3}$





not convex

more examples

① $a \in \mathbb{R}^n,\ b \in \mathbb{R}$

$A = \{x \in \mathbb{R}^n : a^T x \le b\}$

$x, y \in A$

$a^T x \le b \quad \times \alpha$

$+\ a^T y \le b \quad \times (1-\alpha)$

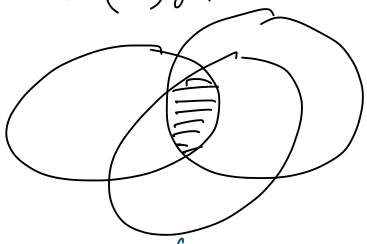
$a^T(\alpha x + (1-\alpha) y) \le \underbrace{\alpha b + (1-\alpha) b}_{= b}$

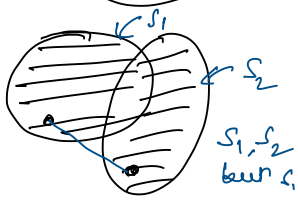
$\Rightarrow \alpha x + (1-\alpha) y \in A$

② $S_1, S_2, \ldots, S_n \subseteq \mathbb{R}^n$ are convex

$S = \bigcap_{i=1}^{n} S_i$

claim: S is convex





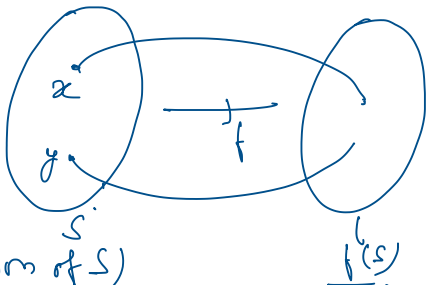
S_1, S_2 convex but $S_1 \cup S_2$ is not

③ $S \subseteq \mathbb{R}^n$ S is convex.

$A \in \mathbb{R}^{m \times n},\ b \in \mathbb{R}^m$

$f : S \to \mathbb{R}^m$

$f(x) = Ax + b$ (affine)



$f(S) = \{Ax + b : x \in S\} \subseteq \mathbb{R}^m$

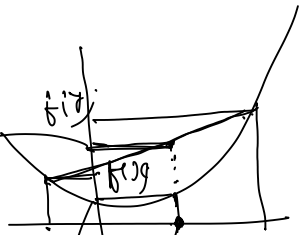
($f(S)$ is called affine transformation of S)

claim: $f(S)$ is convex.

(Exercise)

convex functions

$\alpha f(x) + (1-\alpha) f(y)$



$\alpha \in [0,1]$

$f: S \to \mathbb{R}$

$x \quad \alpha x + (1-\alpha) y \quad y$

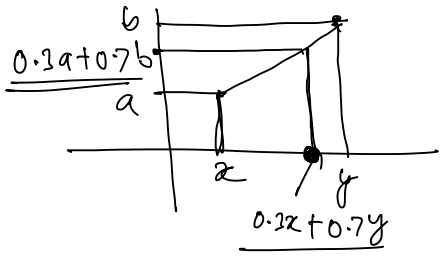
$f(\alpha x + (1-\alpha) y)$

If $\forall x, y \in S$, $\forall \alpha \in [0,1]$

$\alpha f(x) + (1-\alpha) f(y) \geq f(\alpha x + (1-\alpha) y)$

then f is called a convex fn

$\in S$ if S is convex.



$S \subseteq \mathbb{R}^n$ is a ~~convex set~~,

- $f: S \to \mathbb{R}$ is called a convex function if

$$f(\alpha x + (1-\alpha) y) \leq \alpha f(x) + (1-\alpha) f(y)$$

~~for all $\alpha \in [0,1]$~~ for all $x, y \in S$.

Examples ① $f: \mathbb{R} \to \mathbb{R}$

$f(x) = x^2$

$f(\alpha x + (1-\alpha) y) = (\alpha x + (1-\alpha) y)^2$

$= \alpha^2 x^2 + (1-\alpha)^2 y + 2\alpha(1-\alpha) xy$

$= \alpha x^2 + (1-\alpha) y^2 - \alpha(1-\alpha) x^2 - \alpha(1-\alpha) y^2 + 2\alpha(1-\alpha) xy$

$= \alpha x^2 + (1-\alpha) y^2 - \alpha(1-\alpha)(x-y)^2$

$\leq \alpha x^2 + (1-\alpha) y^2$ (if $x \neq y$ & $\alpha \notin \{0,1\}$)

$= \alpha f(x) + (1-\alpha) f(y)$. $\Rightarrow$ strict inequality

$\Rightarrow f$ is convex.

② $f: \mathbb{R} \to \mathbb{R}$

$f(x) = x^4$

$f(\alpha x + (1-\alpha) y) = (\alpha x + (1-\alpha) y)^4$

$= ((\alpha x + (1-\alpha) y)^2)^2$

$\leq (\alpha x^2 + (1-\alpha) y^2)^2$ $\quad$ $(\alpha a + (1-\alpha) b)^2 \leq \alpha a^2 + (1-\alpha) b^2$

strict inequality if $x \neq y$ and $\alpha \notin \{0,1\}$

$\leq \alpha x^4 + (1-\alpha) y^4$

$= \alpha f(x) + (1-\alpha) f(y)$

$\Rightarrow x^4$ is also convex.

$g_1(x) < g_2(x)$

$\geq g_3(x)$

$\Rightarrow g_1(x) < g_3(x)$

③ $f: \mathbb{R}^n \to \mathbb{R}$ $\quad$ $a \in \mathbb{R}^n, b \in \mathbb{R}$

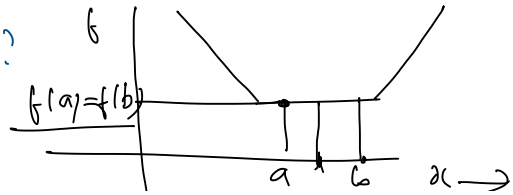
$f(x) = a^T x + b$

claim $f(\alpha x + (1-\alpha) y) = \alpha f(x) + (1-\alpha) f(y)$

$\Rightarrow$ affine functions are convex.

④

Is f convex?



$f(\alpha a+(1-\alpha)b) = \alpha f(a)+(1-\alpha)f(b)$
(equality is allowed)

convexity

$x, y \in S$, $\alpha \in [0,1]$ $f(\alpha x+(1-\alpha)y) \le \alpha f(x)+(1-\alpha)f(y)$

$x=y$, or $\alpha \in \{0,1\} \Rightarrow$ two sides are equal

- $f: S \to \mathbb{R}$ is called a strictly convex function if

$$f(\alpha x+(1-\alpha)y) < \alpha f(x)+(1-\alpha)f(y)$$

for all $\alpha \in (0,1)$ and for all $x, y \in S$ where $y \neq x$.

Examples ① $f(x)=x^2$ is strictly convex.
② $f(x)=x^4$ is strictly convex.
③ Affine functions are not strictly convex.

* $\forall x, y \in S$, $\alpha \in [0,1]$

$$f(\alpha x+(1-\alpha)y) \le \alpha f(x)+(1-\alpha)f(y) - \underbrace{\frac{m}{2}\alpha(1-\alpha)\|x-y\|^2}_{>0 \text{ if } x \neq y,\ \alpha \in \{0,1\}}$$

- $f: S \to \mathbb{R}$ is called a strongly convex function with parameter $m > 0$ if

$$f(\alpha x+(1-\alpha)y) \le \alpha f(x)+(1-\alpha)f(y) - \frac{m}{2}\alpha(1-\alpha)\|x-y\|^2$$

for all $\alpha \in (0,1)$ and for all $x, y \in S$ where $y \neq x$.
α and for all $x, y \in S$

strong convexity $\Rightarrow$ strict convexity $\Rightarrow$ convexity

$\alpha \notin \{0,1\}$
$x \neq y$

$$f(\alpha x+(1-\alpha)y) = \alpha f(x)+(1-\alpha)f(y) - \underbrace{\frac{m}{2}\alpha(1-\alpha)\|x-y\|^2}_{>0}$$

$$\Rightarrow f(\alpha x+(1-\alpha)y) < \alpha f(x)+(1-\alpha)f(y)$$

Ex. ① $f(x)=x^2$

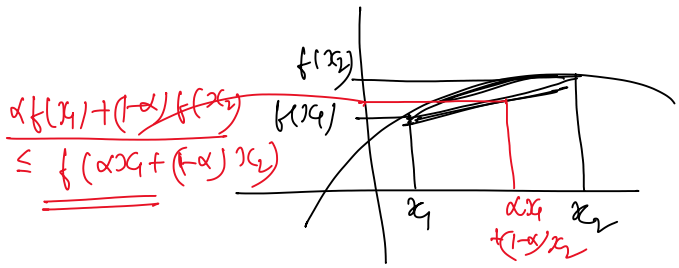
$\frac{m}{2}=1 \Rightarrow m=2$

$$(\alpha x+(1-\alpha)y)^2 = \alpha x^2+(1-\alpha)y^2 - \alpha(1-\alpha)(x-y)^2$$

$\Rightarrow x^2$ is strongly convex with parameter 2

② $f(x)=x^4$
— strictly convex
— but not strongly convex.

$\min_{x \in S} f(x)$
x^*



$S \subseteq \mathbb{R}^n$ is a convex set,

- $f: S \to \mathbb{R}$ is called a concave function if

$$f(\alpha x+(1-\alpha)y) \ge \alpha f(x)+(1-\alpha)f(y)$$

for all $\alpha \in [0,1]$ for all $x, y \in S$.

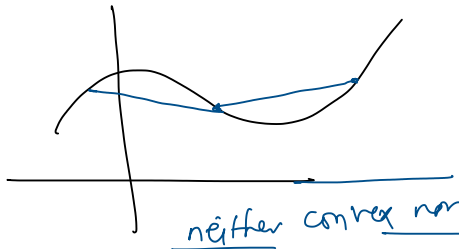
- Can similarly define strictly concave and strongly concave functions

- $f: S \to \mathbb{R}$ concave/strictly concave/strongly concave iff $-f$ convex/strictly convex/strongly convex

f is strongly concave with parameter m

$$f(\alpha x + (1-\alpha) y) \geq \alpha f(x) + (1-\alpha) f(y) + \frac{m}{2}\alpha(1-\alpha)\|x-y\|^2$$

Examples ①



neither convex nor concave.

② $f: \mathbb{R} \to \mathbb{R}$

$f(x) = a^T x + b$ affine

(it is also concave)

③ $f: \mathbb{R}^n \to \mathbb{R}$

$f(x) = \|x\|$?

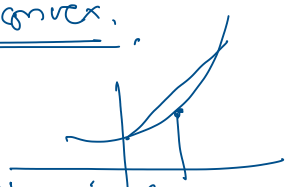
$$f(\alpha x + (1-\alpha) y) = \|\alpha x + (1-\alpha) y\|$$
$$\leq \|\alpha x\| + \|(1-\alpha) y\|$$
$$= \alpha \|x\| + (1-\alpha)\|y\|$$
$$= \alpha f(x) + (1-\alpha) f(y)$$

$\Rightarrow f$ is convex.

$y = 2x$

$\Rightarrow$ equality for any α

$\Rightarrow f$ is not strictly convex.



Is it strictly convex? no.